

Chat Core IQ (dCQ) - Comprehensive Demo Guide

Version: 1.0.2 **Last Updated:** 2026-01-22 **Product:** Chat Core IQ - AI-Powered Citizen Services Platform

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Quick Start URLs

Environment	URL
Production Homepage	https://chat-core-iq.vercel.app/dcq/Home/index.html
Production Admin	https://chat-core-iq.vercel.app/dcq/admin
IVR Demo	https://chat-core-iq.vercel.app/dcq/demo/ivr
Local Homepage	http://localhost:3002/dcq/Home/index.html
Local Admin	http://localhost:3002/dcq/admin

1. Homepage Demo

1.1 Landing Page Overview

When you visit the homepage, you'll see three main components:

Component	Location	Description
Announcements Banner	Top of page	Rotating announcements with navigation arrows
AI Chatbot	Right side	Multi-language chat interface (EN/ES/HT)
FAQ Widget	Below chat	Expandable FAQ accordion

1.2 Demo Script: First-Time User Experience

Step 1: Observe the Announcements

- Notice the banner at the top showing current announcements
- Click the left/right arrows to navigate between announcements
- Close individual announcements with the X button

Step 2: Explore the FAQ Widget

- Scroll down to see the FAQ accordion
- Click any question to expand the answer
- Use the search box to filter FAQs
- Notice multi-language support (EN/ES/HT tabs)

Step 3: Start a Chat Conversation

- Click on the chatbot widget to open
- Select your preferred language (English, Spanish, or Haitian Creole)
- Type a question or select from suggested prompts
- Observe the AI response with source citations

2. AI Chatbot Scenarios

2.1 Language Selection Demo

English Demo:

```
User: "What are your office hours?"
Bot: "Our office hours are Monday–Friday 8:00 AM to 5:00 PM. We are closed on weekends
and federal holidays."
[Sources: Office Information – knowledge base]
```

Spanish Demo:

```
User: "¿Cuáles son sus horarios de oficina?"
Bot: "Nuestro horario de oficina es de lunes a viernes de 8:00 AM a 5:00 PM. Estamos
cerrados los fines de semana y días festivos federales."
```

Haitian Creole Demo:

```
User: "Ki lè biwo ou ouvè?"
Bot: "Biwo nou ouvri Lendi–Vandredi 8:00 AM a 5:00 PM. Nou fèmen nan wikenn ak jou
ferye federal."
```

2.2 Information Inquiry Scenarios

Scenario A: Building Permits

```
User: "How do I apply for a building permit?"
Expected Response: Step-by-step process including:
– Required documents
– Application forms
– Fee information
– Processing timeline
– Online submission options
```

Scenario B: Business License

```
User: "I want to start a new business. What licenses do I need?"
Expected Response: Information about:
– Business license types
```

- Registration requirements
- Zoning compliance
- Home-based business rules
- Fee schedule

Scenario C: Utility Services

User: "How do I set up water service for my new home?"

Expected Response: Details on:

- New account setup
- Required documentation
- Deposit requirements
- Service activation timeline

Scenario D: Payment Questions

User: "Can I pay my property taxes online?"

Expected Response: Information about:

- Online payment portal
- Accepted payment methods
- Payment deadlines
- Installment options

2.3 Escalation Scenarios

When the chatbot cannot answer a question:

User: "I have a complaint about my neighbor's fence"

Bot: "I understand you have a concern about a neighbor's fence. This may require human assistance. Would you like me to:

1. Connect you with Code Enforcement
2. Provide the phone number for our call center
3. Create a service request ticket"

2.4 Feedback Flow

After each response:

1. User sees thumbs up/down buttons
2. Clicking thumbs down shows feedback form
3. Options: "Not Helpful", "Inaccurate", "Need Human Help"
4. Feedback is logged for quality improvement

3. IVR Demo

3.1 Accessing the IVR Demo

URL: <https://chat-core-iq.vercel.app/dcq/demo/ivr>

3.2 IVR Demo Walkthrough

Initial Screen:

- Phone interface simulation
- Keypad (0-9, *, #)
- Language selection at top (EN/ES/HT)

Demo Flow:

- 1. **Welcome Message** (plays automatically)
 - "Welcome to City Services. For English, press 1. Para español, oprima 2. Pou Kreyòl, peze 3."
- 2. **Main Menu (English):**
 - Press 1: Building & Permits
 - Press 2: Business Services
 - Press 3: Utility Services
 - Press 4: Payments
 - Press 5: General Information
 - Press 0: Speak with Representative
- 3. **Sub-Menu Example (Press 1 - Building & Permits):**
 - Press 1: Permit Status
 - Press 2: New Permit Application
 - Press 3: Inspection Scheduling
 - Press 4: Permit Fees
 - Press *: Return to Main Menu

3.3 IVR Demo Features

Feature	How to Test
Keypad Input	Click any number, observe visual feedback
Voice Prompts	Listen to text-to-speech responses
Language Switch	Click EN/ES/HT tabs at top
Transfer Code	Complete a flow to receive unique code (e.g., EYJJIJ)
Menu Navigation	Press * to go back, # to repeat
Call End	Red phone button to end simulation

3.4 Sample IVR Scenarios

Scenario 1: Check Permit Status

```
1 → (Building & Permits)
1 → (Permit Status)
Enter permit number: 12345#
"Your permit BP-12345 is currently APPROVED. Inspection scheduled for January 25, 2026."
```

Scenario 2: Transfer to Representative

0 → (Speak with Representative)
"Please hold while we connect you. Your reference code is: XYZABC"
[Transfer code displayed for reference]

4. Admin Panel Walkthrough

4.1 Admin Dashboard (/dcq/admin)

KPI Overview Cards:

Card	Metric	Description
Total Conversations	1,234	All-time chat sessions
Resolved Today	89%	Questions answered without escalation
Pending	12	Awaiting response or follow-up
Avg Wait Time	2.3s	Response time

Dashboard Sections:

1. Active Now Widget

- Current live sessions
- Queue depth
- Longest wait time

2. Channel Health (6 channels)

- Web Chat:  Online
- IVR:  Online
- SMS:  Online
- Facebook:  Online
- Instagram:  Online
- WhatsApp:  Online

3. Response Time Distribution (Pie Chart)

- Under 5s: 65%
- 5-15s: 25%
- Over 15s: 10%

4. Today vs Yesterday Comparison

- Conversations: +12%
- Resolution Rate: +5%
- Avg Duration: -8%
- Escalations: -15%

Actions:

- Export button (JSON/CSV)
- Refresh button
- Quick links to Escalations and Audit Logs

4.2 Analytics (/dcq/admin/analytics)

Date Range Filters:

- 7 days | 30 days | 90 days | Custom

Available Charts:

1. **Conversation Trend** - Area chart showing daily volume
2. **Peak Hours Heatmap** - 7 days x 8 time slots showing busy periods
3. **Language Distribution** - Pie chart (EN 70%, ES 25%, HT 5%)
4. **Sentiment Analysis** - Positive/Neutral/Negative breakdown
5. **Top Categories** - Bar chart of most common topics
6. **Channel Performance** - Satisfaction % by channel
7. **Resolution Time** - Distribution buckets
8. **Top Questions** - Most frequently asked questions

Export Options:

- Power BI compatible JSON export
- CSV download

4.3 Workflows (/dcq/admin/workflows)

Workflow Types:

Workflow	Status	Description
Appointments	Active	Schedule building permit appointments
Service Requests	Active	Process citizen service tickets
FAQ Actions	Active	Automate FAQ-triggered actions

Appointments Sub-page:

Configure appointment types:

- Building Permit Review (30 min, max 4/slot)
- Business License Consultation (45 min, max 2/slot)
- Code Compliance Meeting (60 min, max 1/slot)
- Zoning Review (45 min, max 2/slot)

Settings per Appointment:

- Duration (minutes)
- Max per time slot
- Lead time (hours)
- Available days
- Time slots

4.4 Content Management (/dcq/admin/content)

Four Content Tabs:

1. **Knowledge Base**
 - 348 knowledge items

- Category filter (8 categories)
- Search functionality
- Bulk edit/delete

2. Custom FAQs

- 7 FAQs with 100% embedding coverage
- CRUD operations
- Priority and status settings
- Language variants

3. Documents

- Upload PDFs, Word docs
- Auto-indexing for search
- Category assignment

4. Web Crawler

- Configure crawl sources
- Schedule frequency
- Indexing status

Content Filters:

- Category (8 options)
- Status (Active, Draft, Archived)
- Priority (High, Medium, Low)
- Language (EN, ES, HT)

4.5 Escalations (/dcq/admin/escalations)

Statistics:

- Total Escalations
- Pending
- In Progress
- Resolved

Escalation Card Details:

- Citizen name and contact
- Escalation reason
- Original chat transcript
- Assigned agent
- Status badges
- Action buttons (Assign, Resolve, Archive)

Filters:

- Status: All | Pending | In Progress | Resolved | Closed
- Search by name, contact, reason

4.6 Notifications (/dcq/admin/notifications)

Filter Tabs:

- All notifications

- Unread
- System alerts
- Activity updates
- Scheduled reminders

Notification Actions:

- Mark as read
- Archive
- Configure preferences

4.7 Announcements (/dcq/admin/announcements)

Management Features:

Action	Description
Create	New announcement with title, body, priority
Edit	Modify existing announcement
Deactivate	Hide from public without deleting
Delete	Permanently remove
Reorder	Drag and drop priority

Announcement Fields:

- Title (EN/ES/HT)
- Body content (rich text)
- Priority (High, Normal, Low)
- Start/End dates
- Target audience

Banner Settings:

- Auto-rotation (on/off)
- Rotation interval (seconds)
- Navigation arrows (show/hide)
- Close button (enable/disable)

4.8 Audit Logs (/dcq/admin/audit-logs)

Log Entry Fields:

- Timestamp
- Admin name
- Action type
- Resource affected
- Location/IP

Action Type Filters:

- Create
- Read
- Update

- Delete
- Login
- Export
- Settings Change

Date Filters:

- Last 24 hours
- Last 7 days
- Last 30 days
- Last 90 days
- Custom range

Security Notice:

- 90-day retention policy displayed
- Export CSV available for compliance

4.9 Settings (/dcq/admin/settings)

Five Settings Tabs:

Profile Tab

- Avatar with initials
- Name, email, phone
- Password change (current, new, confirm)
- Two-factor authentication toggle
- Session timeout setting

Team Tab

Member	Role	Status
Admin User	Admin	Active
Editor 1	Editor	Active
Editor 2	Editor	Inactive
Viewer 1	Viewer	Active
Viewer 2	Viewer	Active

- Invite new member
- Change roles
- Remove access

Permissions Tab

- Three roles: Admin, Editor, Viewer
- Permission matrix (7 resources x 4 actions)
- CRUD toggles per role/resource
- Reset to defaults

Integrations Tab (19+ integrations)

Category	Integrations
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CRM	Salesforce, Microsoft Dynamics 365
SharePoint	3 site connections
IVR	Twilio, Vonage, Amazon Connect
SMS	Twilio, MessageBird, Vonage
Social	Facebook Messenger, WhatsApp Business, Instagram DM
Tyler Technologies	Permitting, 311, Assets, Parks, Munis, Payments, Cashiering, Public Safety, Civic Portal, Content Manager, Records, Data & Insights

Each integration shows:

- Connection status (Connected/Disconnected)
- Last sync timestamp
- Configure button
- Test connection button

Chatbot Tab (8 sub-tabs)

- General settings
- Welcome messages (EN/ES/HT)
- Primary LLM: Claude claude-sonnet-4-20250514
- Backup LLM: GPT-4o-mini
- Temperature: 0.7
- Max tokens: 1024
- Response length
- Personality settings

5. Feature-by-Feature Guide

5.1 Multi-Language Support

Supported Languages:

Code	Language	Coverage
EN	English	100%
ES	Spanish	100%
HT	Haitian Creole	100%

Language Detection:

- Auto-detect from browser
- Manual selection available
- Persists across sessions

5.2 Semantic Search

How It Works:

1. User types question

- 2. Question converted to vector embedding
- 3. pgvector finds semantically similar knowledge items
- 4. Top results sent to Claude/GPT for response generation
- 5. Sources displayed with response

Coverage:

- 348 knowledge items indexed
- 100% embedding coverage
- Real-time search < 200ms

5.3 AI Response Generation

Primary LLM: Claude claude-sonnet-4-20250514 Failover LLM: GPT-4o-mini

Response Features:

- Context-aware answers
- Source citations
- Follow-up suggestions
- Escalation detection

5.4 Feedback System

Feedback Flow:

- 1. User receives response
- 2. Thumbs up/down displayed
- 3. Negative feedback triggers form
- 4. Form options: Not Helpful, Inaccurate, Need Human
- 5. Feedback logged with conversation ID
- 6. Admins review in Analytics

5.5 Export Capabilities

Export Type	Format	Available From
Dashboard Data	JSON, CSV	Dashboard page
Analytics	JSON (Power BI)	Analytics page
Audit Logs	CSV	Audit Logs page
Conversations	JSON	(Future feature)

6. Sample Questions by Category

6.1 Building & Permits

"How do I apply for a building permit?"
"What is the status of my permit application?"
"How long does permit approval take?"
"What documents do I need for a renovation permit?"
"Can I schedule a building inspection online?"
"What are the permit fees for a new construction?"

"Do I need a permit for a fence?"
"How do I get a demolition permit?"

6.2 Business Services

"How do I register a new business?"
"What licenses do I need to open a restaurant?"
"How do I renew my business license?"
"Can I operate a business from my home?"
"What are the zoning requirements for a retail store?"
"How do I apply for a vendor permit?"
"What is the cost of a business license?"
"How do I change my business address?"

6.3 Utility Services

"How do I start water service?"
"Can I pay my water bill online?"
"How do I report a water main break?"
"What are the rates for water service?"
"How do I request a meter reading?"
"Can I set up automatic payments?"
"How do I stop service when I move?"
"What is the process for new construction water hookup?"

6.4 Payments & Taxes

"Can I pay my property taxes online?"
"When are property taxes due?"
"How do I get a copy of my tax bill?"
"What payment methods do you accept?"
"Is there a payment plan for taxes?"
"How do I appeal my property assessment?"
"Where can I pay in person?"
"Do you accept credit cards?"

6.5 General Information

"What are your office hours?"
"Where is the main office located?"
"How do I contact customer service?"
"Is there parking available?"
"Can I make an appointment?"
"What holidays are you closed?"
"How do I file a complaint?"
"Do you have wheelchair accessibility?"

6.6 Code Enforcement

"How do I report a code violation?"
"What constitutes a property violation?"
"How do I appeal a code violation notice?"
"What are the fines for code violations?"
"How do I check if my property is in compliance?"
"Can I report a violation anonymously?"
"What is the timeline for violation resolution?"
"Who do I contact about a noise complaint?"

7. End-to-End Workflows

7.1 Citizen Journey: Applying for a Building Permit

Step 1: Initial Inquiry (Chatbot)

Citizen: "I want to build an addition to my house. Where do I start?"
Bot: "Great! To build an addition, you'll need a Building Permit. Here's the process:
1. Submit application form
2. Include architectural drawings
3. Pay permit fee (\$150-\$500 based on value)
4. Wait for plan review (5-10 business days)
5. Schedule inspection after approval

Would you like me to:
A) Provide the application form link
B) Explain required documents
C) Help schedule a consultation"

Step 2: Follow-up Questions

Citizen: "What documents do I need?"
Bot: "For a residential addition permit, you'll need:
- Site plan showing property boundaries
- Floor plans with dimensions
- Structural drawings (if applicable)
- Energy compliance forms
- Contractor license (if using contractor)
- Property survey (for setback verification)

[Source: Building Permit Requirements – Knowledge Base]"

Step 3: Appointment Scheduling

Citizen: "Can I schedule a consultation?"
Bot: "I can help you schedule a Building Permit Consultation. Available slots:
- Monday 1/23: 10:00 AM, 2:00 PM
- Tuesday 1/24: 9:00 AM, 11:00 AM, 3:00 PM
- Wednesday 1/25: 10:00 AM, 2:00 PM

Which time works best for you?"

Step 4: IVR Follow-up

Citizen calls 311 → Press 1 (Building & Permits) → Press 3 (Inspection Scheduling)
"Enter your permit number followed by pound..."
"Your inspection for permit BP-2024-123 is scheduled for January 25 at 10:00 AM."

7.2 Admin Workflow: Managing Knowledge Base

Step 1: Identify Gap

- Review Analytics → Top Questions
- Notice "Solar panel permits" trending but low resolution rate

Step 2: Create Content

- Go to Content → Knowledge Base
- Click "Add Item"
- Fill in:
 - Title: "Solar Panel Permit Requirements"
 - Category: Building & Permits
 - Content: Detailed requirements
 - Languages: EN, ES, HT
- Save → Auto-generates embedding

Step 3: Verify

- Test in chatbot: "How do I get a solar panel permit?"
- Confirm accurate response with new source

Step 4: Monitor

- Check Analytics after 24 hours
- Verify resolution rate improved

7.3 Admin Workflow: Handling Escalations

Step 1: Review Escalation

- Dashboard shows "2 Pending Escalations"
- Click to view Escalations page
- Select escalation: "Citizen upset about permit denial"

Step 2: Review Context

- Read original chat transcript
- Note: Citizen asked about permit, got info, but wanted appeal process
- Bot couldn't provide specific appeal procedures

Step 3: Take Action

- Click "Assign to Agent"
- Select: Building Department supervisor
- Add note: "Citizen needs appeal process guidance"
- Status: In Progress

Step 4: Follow-up

- Agent contacts citizen

- Resolves issue
- Updates status: Resolved
- Adds resolution note

Step 5: Improve

- Add "permit appeal process" to Knowledge Base
- Future inquiries handled automatically

7.4 Multi-Channel Journey

Scenario: Citizen starts on web, continues on phone

Web Chat (10:00 AM):

Citizen: "I need to report a pothole on Main Street"

Bot: "I can help you report a pothole. I'll need:

- Exact location (address or cross streets)
- Size estimate (small, medium, large)
- Any safety concerns

What is the location?"

Citizen: "Corner of Main and 5th Street, it's pretty large"

Bot: "Thank you! I've created Service Request #SR-2024-456 for:

- Location: Main St & 5th St intersection
- Issue: Large pothole
- Priority: High (traffic intersection)

Estimated repair: 2-3 business days

Reference number: SR-2024-456

Would you like text updates on this request?"

Phone IVR (2:00 PM):

Citizen calls → Press 5 (General Information) → Press 2 (Service Request Status)

"Please enter your service request number followed by pound: 2024456#"

"Service Request SR-2024-456 for pothole repair at Main and 5th Street is currently IN PROGRESS. A repair crew has been dispatched. Estimated completion: today by 5:00 PM."

7.5 Announcement Management Workflow

Step 1: Create Emergency Announcement

- Admin → Announcements → Create New
- Title: "Water Main Break - North District"
- Body: "Emergency water main repair in progress. North District residents may experience low pressure. Estimated restoration: 6:00 PM today."
- Priority: High
- Languages: EN, ES, HT
- Activate immediately

Step 2: Configure Banner

- Banner Settings → Enable auto-rotation OFF (emergency static)
- Set as first position

Step 3: Monitor

- Dashboard shows announcement views
- Chatbot includes announcement context

Step 4: Update

- Service restored
 - Edit announcement: "RESOLVED - Water service restored..."
 - Change priority: Normal
 - Schedule deactivation: tomorrow 8:00 AM
-

Appendix: Quick Reference Cards

Chatbot Commands

Action	How
Change language	Click language button (EN/ES/HT)
Start new chat	Click "New Conversation"
Give feedback	Click thumbs up/down
Request human	Say "talk to someone"

IVR Navigation

Key	Action
1-9	Menu selection
0	Speak to representative
*	Go back
#	Repeat/Confirm
Red phone	End call

Admin Shortcuts

Page	URL Path
Dashboard	/dcq/admin
Analytics	/dcq/admin/analytics
Workflows	/dcq/admin/workflows
Content	/dcq/admin/content
Escalations	/dcq/admin/escalations

Notifications	/dcq/admin/notifications
Announcements	/dcq/admin/announcements
Audit Logs	/dcq/admin/audit-logs
Settings	/dcq/admin/settings